

Week 12

Option A → piecewise auxiliary model

Instead of one constant `P_aux`, use a low-dimensional dynamic model

$$P_{aux}(t) = P_{base} + P_{door} I_{door}(t) + P_{stop} I_{stop}(t) + P_{hvac} I_{season}(t)$$

- `P_base` = always-on auxiliaries
- `I_door(t)` = 1 when doors are open, else 0
- `I_stop(t)` = 1 when bus is stopped / near stopped
- `I_season(t)` = winter/summer proxy for HVAC load

Then maybe clip to whatever interval we want

$$P_{aux}(t) = \text{clip}(P_{aux}(t), 2, 8)$$

Option B

One `P_aux` for each bus and season

- `P_aux_bus183_winter`
- `P_aux_bus183_summer`
- `P_aux_bus208_winter`
- `P_aux_bus208_summer`

HVAC variation appears indirectly

Option C → per-bus / per-season constant

estimate auxiliary demand from time periods where tractive demand is small:

- bus stopped
- very low speed
- near-zero acceleration
- flat road
- no strong propulsion or braking demand

$$P_{measured}(t) \approx P_{aux}(t)$$

Then build an empirical auxiliary model from these segments.

Use selected segments to fit something like

$$P_{aux}(t) = \beta_0 + \beta_1 I_{door}(t) + \beta_2 I_{stop}(t) + \beta_3 I_{winter}(t)$$

Then use `P_aux(t)` everywhere in the energy model